

Best practices for driving high power parallel MOSFETs in a half-bridge configuration

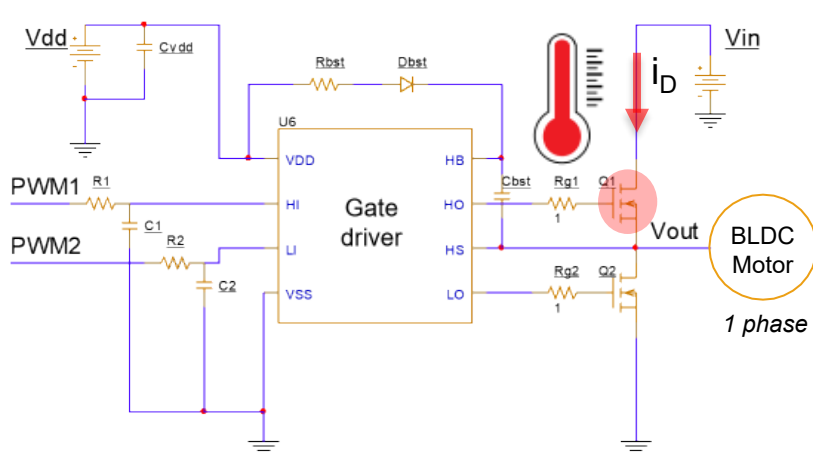
Leslie Marquez Arroyo

Session Objectives

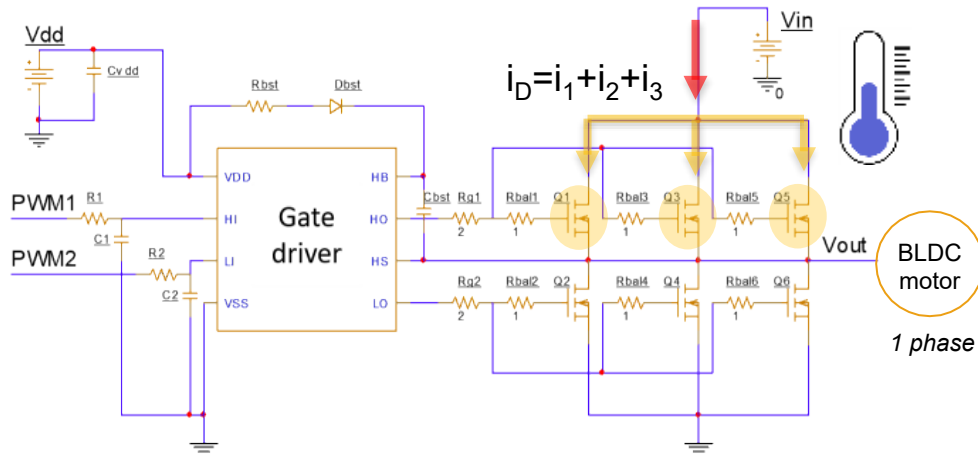
Provide a deeper understanding of:

- When are parallel MOSFETs desired?
- What are the main challenges?
- Best design practices
 - Gate driver and FET selection
 - Schematic design
 - Layout design
- Debugging tips

Parallel FETs enable higher power applications

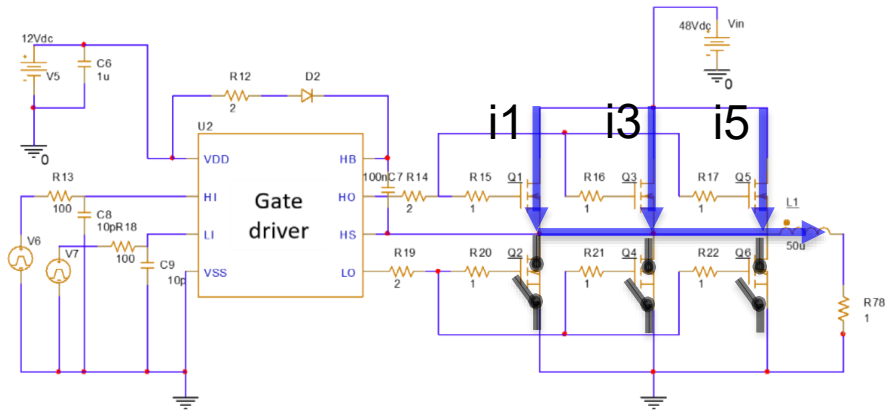


*Motor drive application
using single half-bridge*

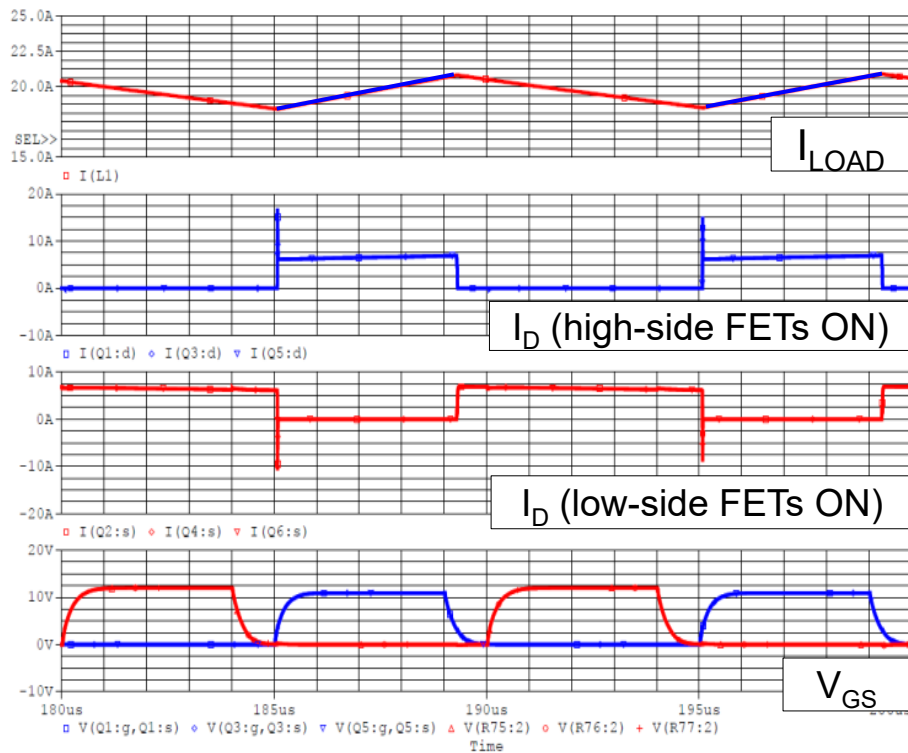


*Motor drive application **using**
3 half-bridges in parallel*

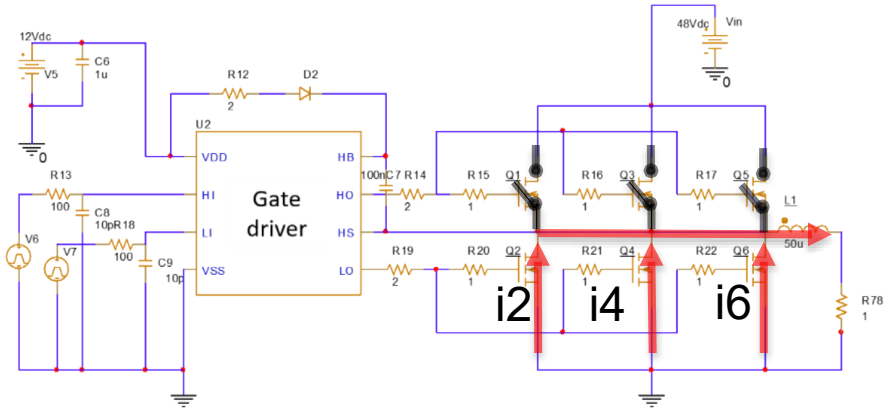
Ideally, current is shared equally between parallel FETs



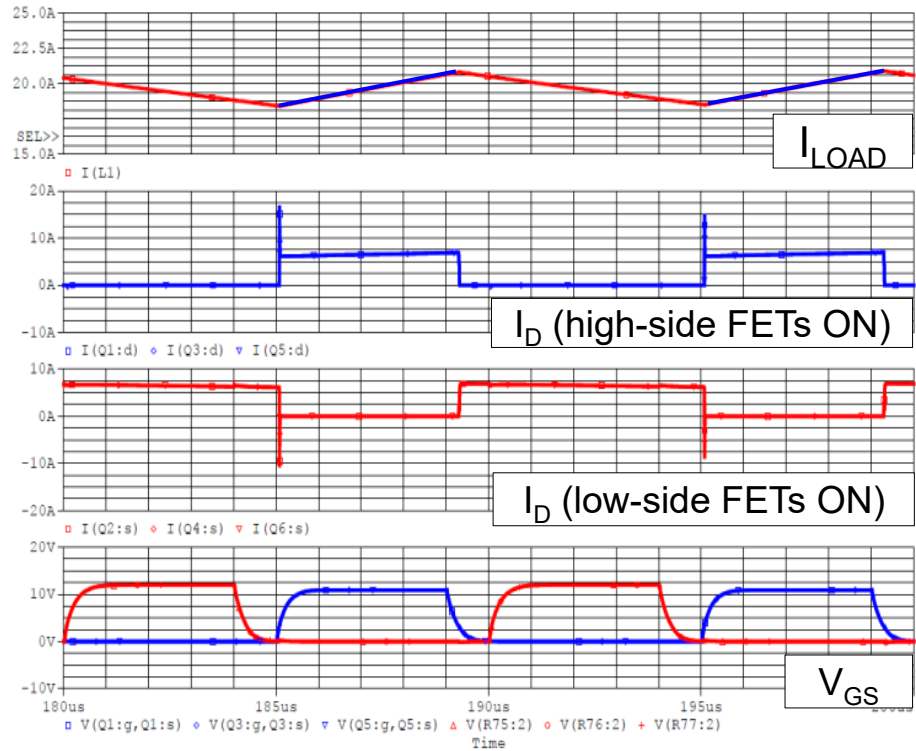
*3 parallel FETs in half bridge configuration:
High-side FETs on*



Ideally, current is shared equally between parallel FETs

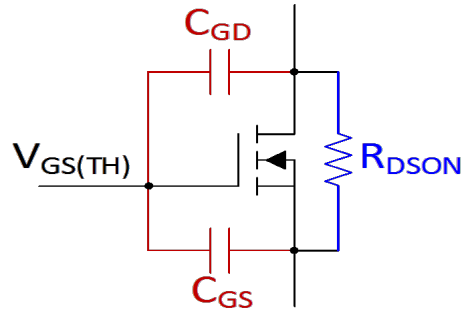


*3 parallel FETs in half bridge configuration:
Low-side FETs on*



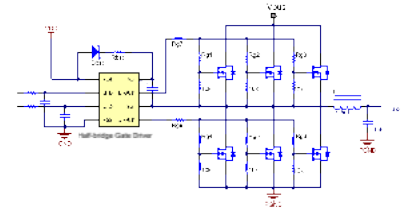
In reality, several factors affect this balance

MOSFET characteristics



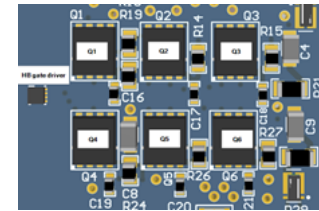
- During transients
- During steady state

Factors external to MOSFET

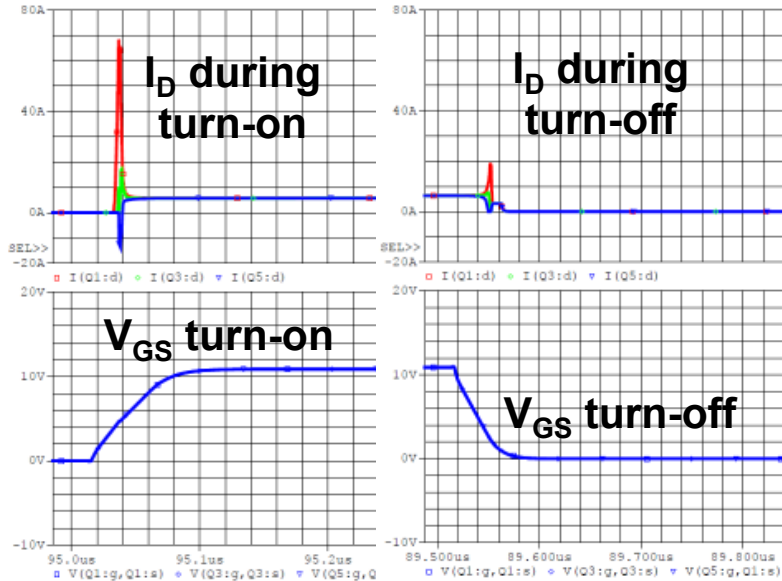


Schematic design

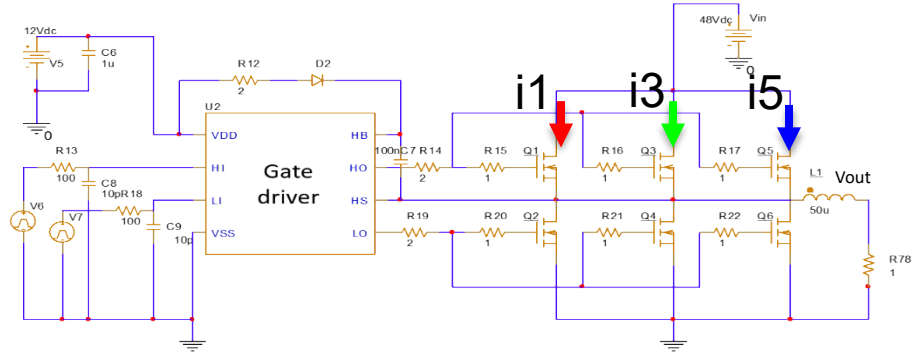
Layout design



$V_{GS(TH)}$ part-to-part variation has the most impact on current imbalance

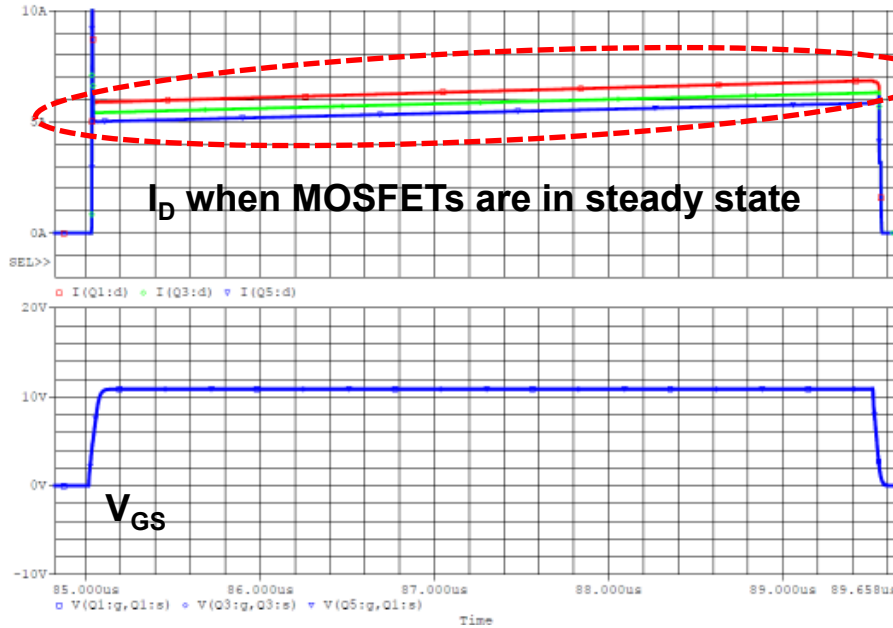


$Q1 V_{GS(TH)} = 2.5V,$
 $Q3 V_{GS(TH)} = 3.0V,$
 $Q5 V_{GS(TH)} = 3.5V$

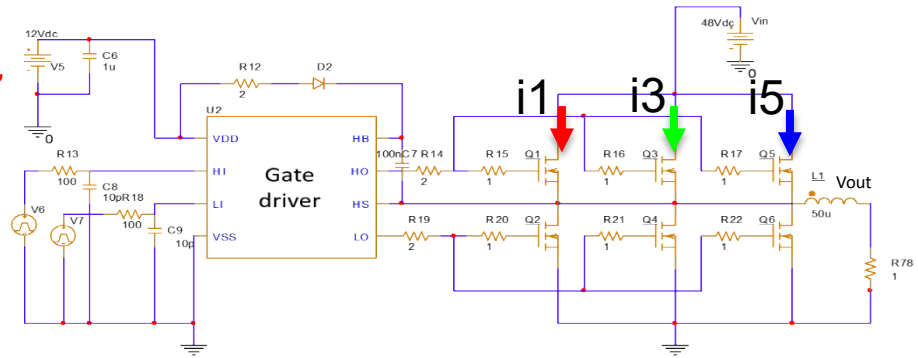


- FET with lowest $V_{GS(TH)}$ switches on 1st, and carries largest amount of current during turn on.
- Higher current causes higher power dissipation on the FET and therefore higher temperature.
- FET's $V_{GS(TH)}$ has negative temperature coefficient, and therefore $V_{GS(TH)}$ will decrease even further.

R_{DSon} part-to-part variation creates current imbalance in steady state



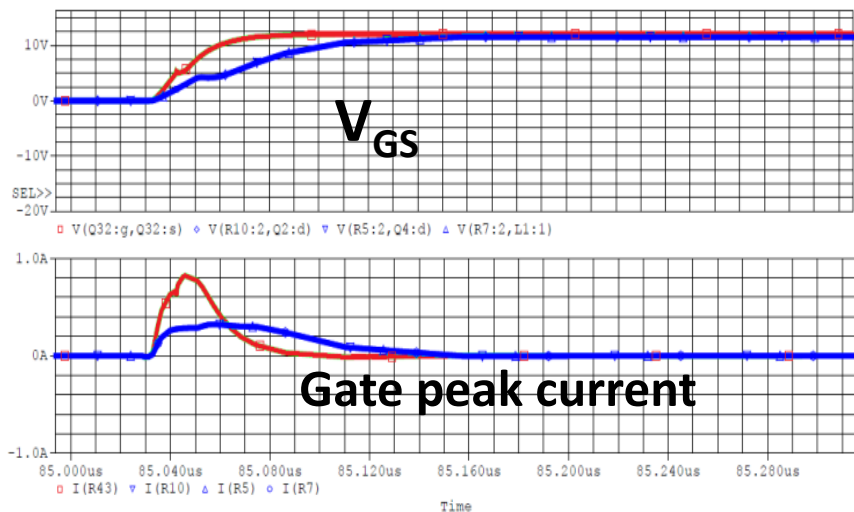
Q1 R_{DSon} = 5.4m Ω ,
Q3 R_{DSon} = 5.9m Ω ,
Q5 R_{DSon} = 6.4m Ω



- FET with lowest R_{DSon} carries higher I_D .
- As a result, the FET dissipates more power and it heats up.
- R_{DSon} has positive temperature coefficient, and therefore the resistance increases, balancing the current between the FETs.

Parasitic capacitances will multiply

Effects of parallel C_{GS}



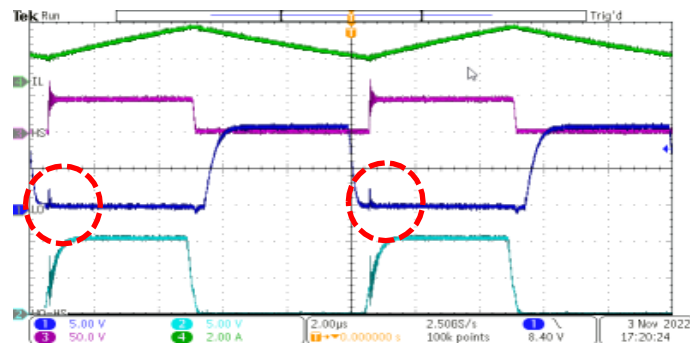
— 3 FETs in parallel
— 1 FET

Effects of parallel C_{GD}

Miller capacitance will multiply:

Current injected $\leftarrow I = C_{GD} * dV/dt \rightarrow$ Switch node
to low-side gate Miller capacitance slew rate

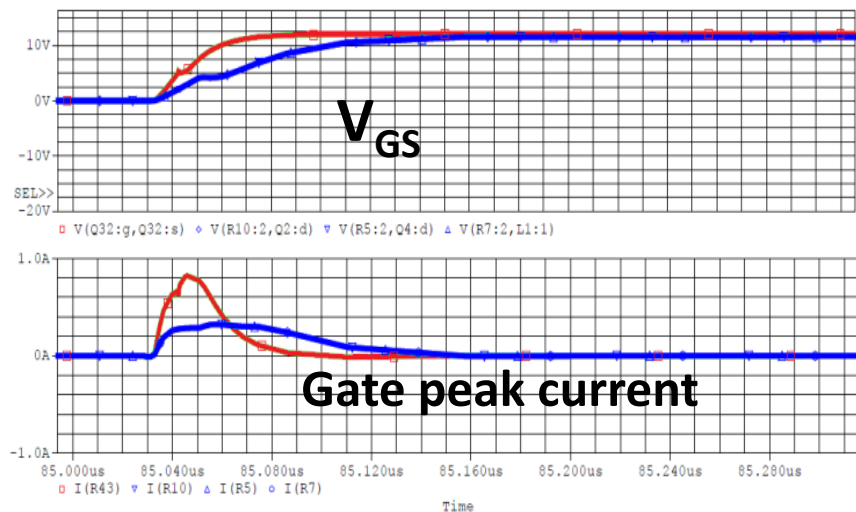
C_{GD}/C_{GS} ratio remains the same: $C_{GD}/C_{GS} = 3C_{GD}/3C_{GS}$



Effects of miller charge injected on LO

Parasitic capacitances will multiply

Effects of parallel C_{GS}



— 3 FETs in parallel
— 1 FET

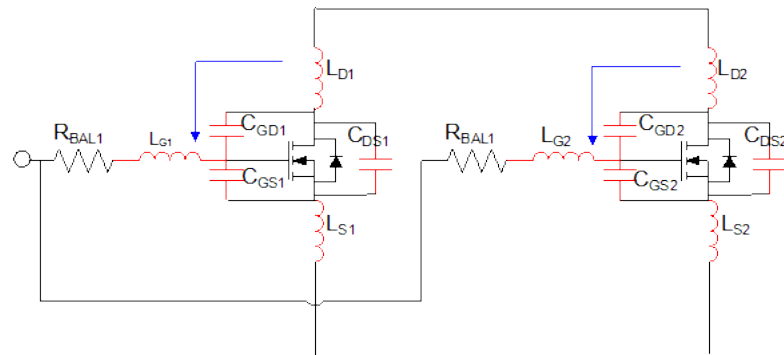
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slew rate

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Parasitic Oscillations



Key care-about for FET selection

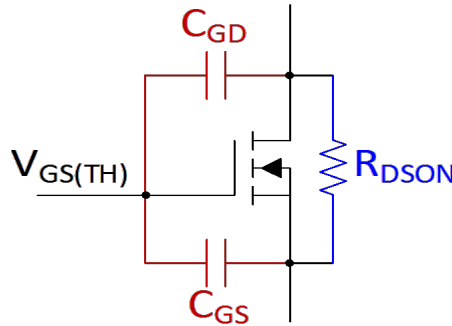
- If FET with power capability need is available, use a single FET.
- If available FET has high $R_{DS(on)}$, paralleling them will help reduce $R_{DS(on)}$.

To minimize miller charge:

- Select a FET with low C_{GD}/C_{GS}
- If not possible and C_{GD} causes low-size FET to turn on, plan for ext. miller clamp

To minimize current imbalance during turn-on/turn-off:

- Use same FET P/N for all
- R_G balancing resistors at gate



$R_{DS(on)}$ has lower impact since its positive temperature coefficient will help balance the current during static operation

Consider C_{GS} when selecting a gate driver to have enough drive strength

Key care-about for gate driver selection

- Estimate drive strength needed:
 - FET's Q_g * number of parallel FETs
 - Turn on/off speed requirements
- Consider maximum voltage at the switch node
- Estimate maximum power that driver can dissipate
- Miller clamp: If high C_{GD} causes undesired FET turn-on, plan for a miller clamp

3 parallel CSD19531Q5A MOSFETs

50ns turn on/off requirement

- $Q_g = 48nC$ (max) @ $V_{gs} = 10V$
- Total gate charge: $48nC * 3 = 144nC$
- Equivalent capacitance assuming 10V V_{gs} : **14.4nF**

$$I_{pk} = C * \frac{dV}{dt} = 14.4nF * \frac{10V}{50ns} = 2.88A$$

*Assumptions: 0Ω gate resistor and $V_{gs}=10V$

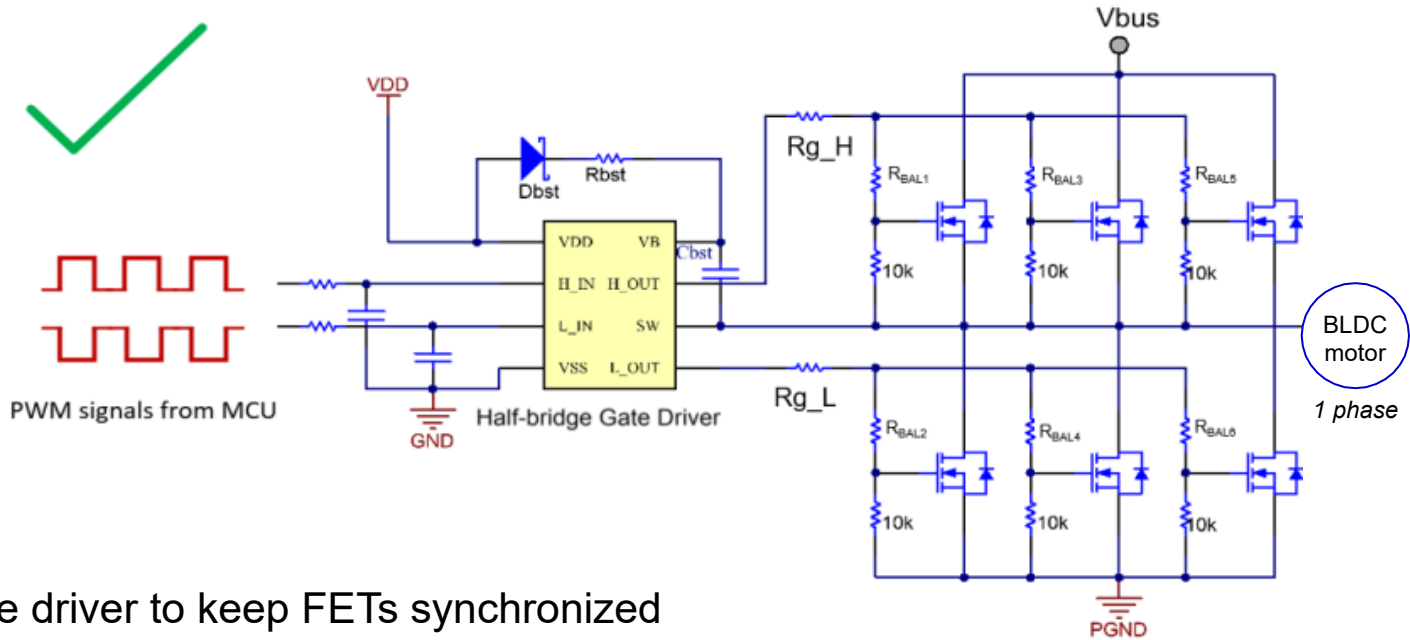
Gate driver options for high voltage

Example of gate drivers' ratings:

Driver solution	Max Vbus	Max drive strength
Non-isolated half-bridge	1.2KV	4A
Isolated dual-channel	2KV	6A
Non-isolated low-side	N/A	10A
Isolated single channel	1.5KV 2KV	30A 17A

For higher voltage and drive strength, non-isolated half bridge drivers available in the market are limited. Consider using an isolated gate driver.

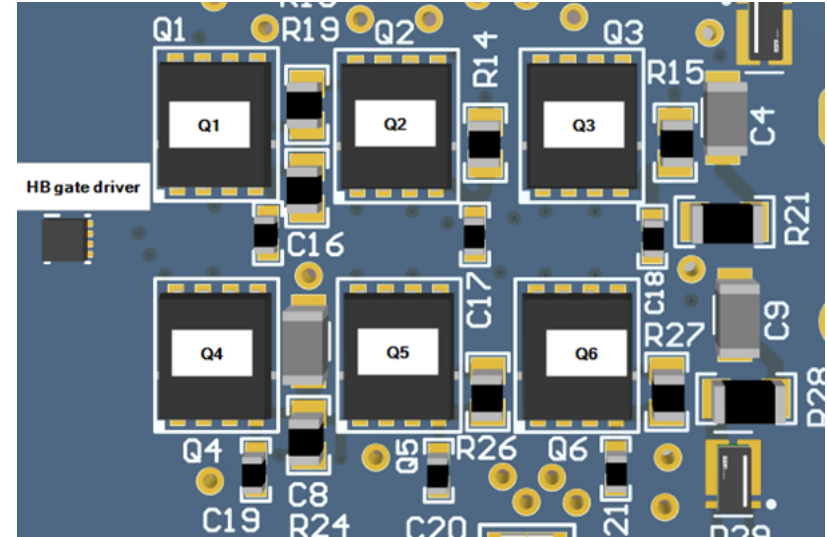
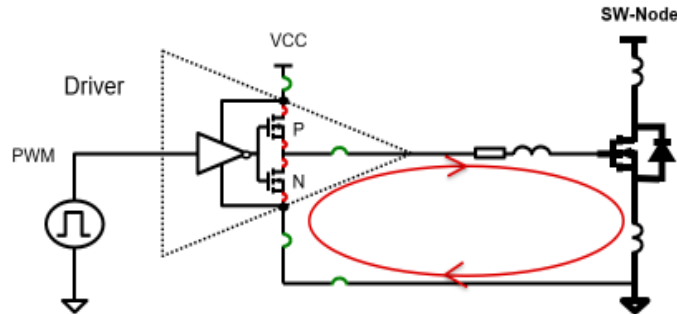
Schematic design recommendations



- ✓ Use single driver to keep FETs synchronized
- ✓ R_G at the driver's output (R_{g_H} , R_{g_L}) sets turn on/turn off speed
- ✓ R_{BAL} at the gate of each FET (R_{BAL1} to R_{BAL6}) are balancing resistors
- ✓ Ok to use $R_{pulldown}$ at each FETs gate (individual to keep FET off)

PCB design recommendations

- ✓ Place R_G of the driver output right at the driver's pin. Place R_{BAL} right at each FET's gate pin
- ✓ Keep traces/placement symmetrical
- ✓ Use copper plane to help with MOSFET's thermals
- ✓ Minimize loop inductance:



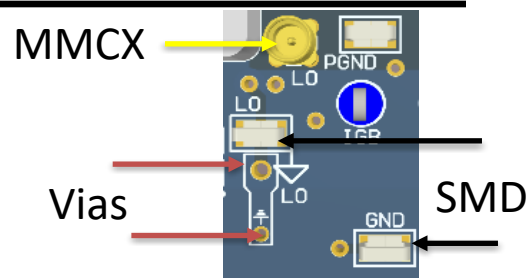
Layout example: Half-bridge configuration using 3 MOSFETs in parallel

Key differences when paralleling half-bridges

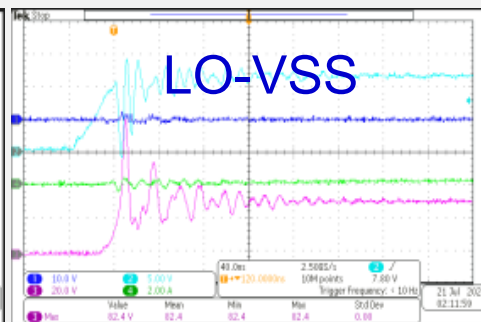
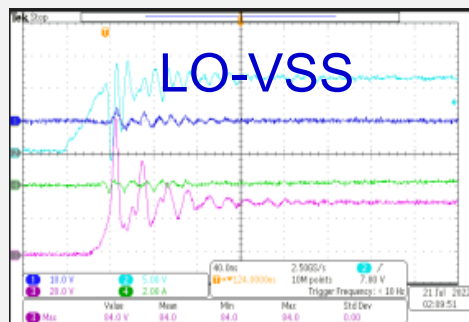
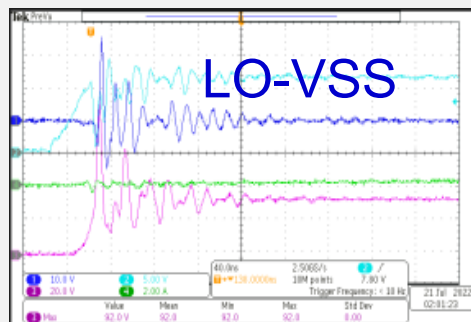
- Symmetricity in layout becomes more challenging.
- Depending on drive strength needed, need to consider half-bridge driver, isolated driver or low-side driver.
- Shoot-through can occur due to increased miller charge.

Debugging tips for testing and evaluation

- Plan for accurate measurements:
 - Test points as close as possible to driver pins and FET pins



Signal:



Test point:

SMD

Vias

MMCX

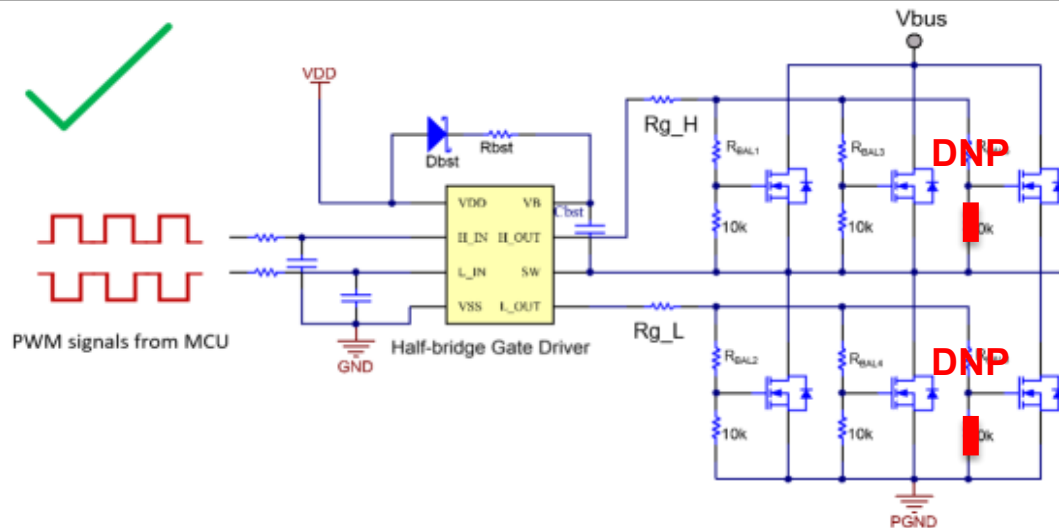
Probe connector:

Probe hook
+ GND lead

Tip and barrel

MMCX

Debugging tips for testing and evaluation



If number of parallel FETs must be decreased for testing, balance resistor at the gate of the MOSFET can be removed. Short gate to source to avoid MOSFET accidentally turning on.

Adjust balance resistors (Rbal1 to Rbal6) to help with parasitics due to layout:

- Capture V_{GS} measured directly at the FETs' gate and source.
- If FET has high ringing (e.g. due to parasitic trace inductance) a large balance resistor is needed to dampen the oscillations.
- Goal is to minimize differences in V_{GS} rise/fall waveforms between FETs.

Takeaways

1. Parallel MOSFETs are used in high power applications
 - Where available FETs are not rated for such high power
 - Where the FET's $R_{DS(on)}$ is larger than desired
2. The balance of the current shared between parallel FETs is affected by several factors
 - Characteristics of the FET
 - Layout and schematic
3. Following best practices for FET selection, driver selection and board design can help minimize the differences in current imbalance between FETs
4. Based on bus voltage and FET turn on/off time needed, select a driver that meets the drive strength needed.

References

- “Tips for Successfully Paralleling Power MOSFETs” application brief: <https://www.ti.com/lit/an/slpa020/slpa020.pdf>
- “Parallel Configuration of H-Bridges” application note: <https://www.nxp.com/docs/en/application-note/AN4833.pdf>
- “Paralleling Power MOSFETs in High Current Applications” application note: https://www.infineon.com/dgdl/Infineon-PowerMOSFET_Paralleling_power_MOSFET_in_high_current_applications-ApplicationNotes-v02_01-EN.pdf?fileId=5546d462758f5bd101759ca434f111c1&da=t
- “MOSFET Paralleling (Parasitic Oscillation between Parallel Power MOSFETs)” applications note: https://toshiba.semicon-storage.com/info/application_note_en_20180726_AKX00067.pdf?did=59458
- “IGBT Gate Driver Reference Design for Parallel IGBTs With Short-Circuit Protection and External BJT Buffer” reference design: <https://www.ti.com/lit/ug/tiduc70a/tiduc70a.pdf>
- “18-V/1-kW, 120-A_{PEAK}, >98% Efficient, High Power Density Brushless Motor Drive Reference Design” application brief: <https://www.ti.com/lit/ug/tiduc10/tiduc10.pdf>
- “A basic brushless date driver design” blog series: https://e2e.ti.com/blogs_/b/industrial_strength/posts/a-basic-brushless-gate-driver-design-part-1
- “CSD19531Q5A 100V N-Channel NexFET™ Power MOSFETs” data sheet: <https://www.ti.com/lit/ds/symlink/csd19531q5a.pdf>
- “Issues on Connecting MOSFETs in Parallel” video by Lewis Loflin: <https://www.youtube.com/watch?v=AOf-i2vLd2o>
- “How to Parallel Power MOSFETs – Quick Learning” video by Nexperia: <https://www.youtube.com/watch?v=0wFdpYmEKsS>
- “Parallel Multiple MOSFETs Using Optimized Current Sharing Technology” video by Nexperia: <https://www.youtube.com/watch?v=830-0kEWa68&t=4s>
- “Paralleling MOSFETs in High Power Applications” video by Nexperia: <https://www.youtube.com/watch?v=BW9853Ak8QM>